

EXTENDED RTABMAB AND REDUCED FAST SCNN FOR AUTONOMOUS INDUSTRIAL VEHICLE NAVIGATION

Azzam Wildan Maulana

Department of Electrical Engineering
Institut Teknologi Sepuluh Nopember
Surabaya, Indonesia 60111
6022241101@student.its.ac.id

Rudy Dikairono

Department of Electrical Engineering
Institut Teknologi Sepuluh Nopember
Surabaya, Indonesia 60111
rudydikairono@its.ac.id

Ahmad Zaini

Department of Electrical Engineering
Department of Computer Engineering
Institut Teknologi Sepuluh Nopember
Surabaya, Indonesia 60111
zaini@its.ac.id

Mauridhi Hery Purnomo

Department of Electrical Engineering
Department of Computer Engineering
Institut Teknologi Sepuluh Nopember
Surabaya, Indonesia 60111
hery@ee.its.ac.id

Moh Ismarintan Zazuli

Department of Electrical Automation Engineering
Institut Teknologi Sepuluh Nopember
Surabaya, Indonesia 60111
ismarintan@its.ac.id

Muhtadin

Department of Computer Engineering
Institut Teknologi Sepuluh Nopember
Surabaya, Indonesia 60111
muhtadin@ee.its.ac.id

Abstract—This research develops a dual-layer navigation system for industrial autonomous vehicles. The first layer, an Extended RTAB-Map, performs global navigation and mapping. The second, a Reduced Fast-SCNN, handles local perception for lane detection. The Extended RTAB-Map introduces a skip connection from hardware odometry to pose fusion. The Reduced Fast-SCNN decreases convolutional channels and replaces pyramid pooling with average pooling. Experiments show an increase in pose update rate from 1 Hz to 50 Hz with centimeter-level accuracy (error < 5 cm). The Reduced Fast-SCNN achieves real-time CPU-only inference below 10 ms.

Index Terms—navigation system, RTAB-Map, Fast-SCNN, graph-based SLAM, machine learning, lane detection, autonomous vehicle

I. INTRODUCTION

Autonomous vehicles are increasingly vital in industrial logistics, offering improvements in safety, efficiency, and operational precision. Industrial plants often require continuous material transport in constrained and hazardous environments, where human operators are prone to fatigue and error. As reported in [1], night-shift workers are particularly susceptible to micro-sleep episodes lasting 1–2 seconds, which can result in serious accidents. To mitigate these risks, autonomous vehicles can replace or assist human-operated systems, ensuring continuous and safe material handling.

An autonomous navigation system generally consists of two main components: the global navigation system for pose

estimation and mapping, and the local navigation system for short-range perception and motion control. The global system requires both accuracy and high update rates to maintain stable control. In this study, we extend the Real-Time Appearance-Based Mapping (RTAB-Map) algorithm [2], a graph-based SLAM framework known for its robustness in visual and LiDAR mapping. However, the default RTAB-Map provides a limited update rate (around 1 Hz) and is heavily dependent on visual odometry, which suffers under lighting variations or occlusions [3]. To overcome these limitations, an *Extended RTAB-Map* is proposed by introducing a skip connection from hardware odometry to pose fusion. Odometry data from encoders and IMU sensors are combined with RTAB-Map's corrected pose using a Kalman Filter [4], improving both update frequency and pose accuracy.

For local navigation, lane or drivable-area detection is essential for maintaining safe motion. Deep learning-based semantic segmentation models like Fast-SCNN [5] are efficient but still computationally heavy for CPU-only systems. Although D-Fast-SCNN [6] improves accuracy using dilation, it increases inference time. To meet real-time industrial demands, we design a *Reduced Fast-SCNN* by lowering convolutional channels and replacing pyramid pooling with average pooling, achieving sub-10 ms inference on CPU. The output segmentation provides a confidence level used to adjust vehicle speed and steering dynamically. By integrating the Extended RTAB-

Map and Reduced Fast-SCNN, this research delivers a high-frequency, CPU-efficient, and centimeter-accurate navigation system suitable for industrial autonomous vehicles.

II. METHODOLOGY

The methodology of this research is divided into three main parts. The first part discusses the architecture of the Extended RTAB-Map. The second part presents the network design of the Reduced Fast-SCNN. The final part explains the integration of the Extended RTAB-Map and Reduced Fast-SCNN into a unified navigation system for autonomous industrial vehicles.

A. Extended RTAB-Map for Global Navigation System

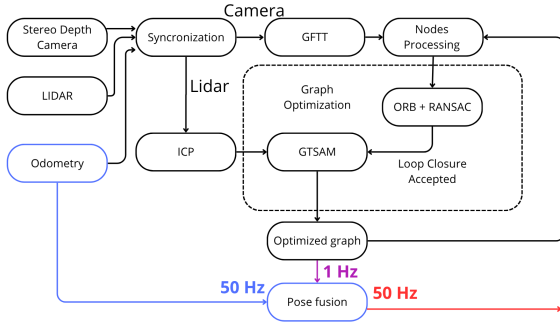


Fig. 1. Block diagram of graph-based SLAM in mapping mode.

The first and most important step is to calculate the hardware odometry using encoder speed and IMU data. The relationship can be expressed as follows:

$$\begin{bmatrix} v_x \\ v_y \\ v_\theta \end{bmatrix} = \frac{1}{\Delta t} \begin{bmatrix} \text{encoder_to_meter} \text{encoder_speed} \cos \theta \\ \text{encoder_to_meter} \text{encoder_speed} \sin \theta \\ \text{current_IMUyaw} - \text{previous_IMUyaw} \end{bmatrix} \quad (1)$$

Here, v_x represents the linear velocity in the x-direction, v_y is the linear velocity in the y-direction, and v_θ is the angular velocity around the z-axis. The term Δt denotes the time difference between the current and previous readings (20 ms). The variable `encoder_to_meter` is a conversion factor from encoder speed (in rpm) to meters per second. `encoder_speed` refers to the encoder output, θ is the yaw angle from the IMU in radians, while `current_IMUyaw` and `previous_IMUyaw` denote the consecutive IMU yaw readings in degrees.

The second step is to utilize and optimize the RTAB-Map framework. The first enhancement involves performing approximate time synchronization among camera, LiDAR, and hardware odometry data. This step is essential because each sensor operates at a different update frequency, and desynchronized data could lead to mapping errors.

The second optimization of RTAB-Map involves the use of the GFTT (Good Features to Track) algorithm as an image feature detector. As demonstrated in [7], GFTT provides a fast yet reliable feature extraction performance suitable for

real-time applications. The third improvement applies the combination of ORB (Oriented FAST and Rotated BRIEF) and RANSAC (Random Sample Consensus) for feature matching. ORB provides a computationally efficient descriptor while maintaining robustness, and RANSAC filters out outliers from matched keypoints [8]. The output of this step is a loop-closure hypothesis, which indicates that the robot has revisited a previously mapped location. This hypothesis is used to refine and optimize the pose graph.

The fourth enhancement in RTAB-Map involves applying the ICP (Iterative Closest Point) algorithm to establish constraints between the nodes. ICP is utilized to compute the relative pose and covariance between two adjacent nodes [9]. This algorithm is selected because LiDAR laser scans provide higher geometric accuracy compared to depth information from cameras.

The fifth stage involves full graph optimization using all nodes and edges. When a valid loop-closure hypothesis is detected, the corrected pose and covariance are calculated based on the ICP constraint. These values are then used to optimize the entire pose graph through nonlinear optimization using the GTSAM library [10]. This optimization reduces accumulated drift and improves global map consistency.

Despite these improvements, the standard RTAB-Map implementation yields an average pose update rate of only 1 Hz when tested on an Intel i5 12th Gen processor with 16 GB of RAM and no dedicated GPU. To increase the update rate, we introduce a skip connection from the hardware odometry to the pose fusion module (indicated by the blue line in Fig. 1). The pose fusion employs a Kalman Filter to combine the corrected pose and covariance from RTAB-Map with the odometry-based pose and covariance from the hardware sensors. The hardware odometry, calculated from Eq. 1, serves as the velocity model in the Kalman Filter, while the RTAB-Map output acts as the measurement model.

The fusion strategy begins by assigning a low covariance value (around 0.01) to the hardware odometry, prioritizing responsiveness during the vehicle's initial motion. As the system detects more loop closures, the covariance of the RTAB-Map corrected pose gradually decreases, allowing it to dominate the fusion process. Consequently, the final pose estimation becomes both accurate and smooth over time. The use of the Kalman Filter not only improves pose accuracy but also minimizes abrupt transitions during map corrections. The resulting fused pose achieves a 50 Hz update rate on the same hardware setup, with centimeter-level precision suitable for real-time navigation.

B. Reduced Fast-SCNN for Local Navigation System

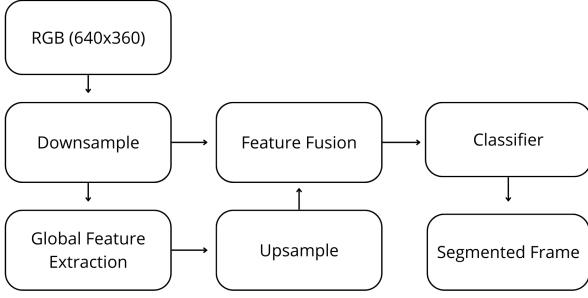


Fig. 2. Block diagram of the Reduced Fast-SCNN architecture.

Building upon the original Fast-SCNN architecture [5], we developed a modified version specifically optimized for lane detection in industrial settings. The original Fast-SCNN was designed for high-speed semantic segmentation on mobile devices; however, in our application, the focus is shifted from segmentation accuracy toward achieving real-time inference on limited computing hardware. The overall structure of the Reduced Fast-SCNN is shown in Fig. 2.

1) *Downsampling*: The Reduced Fast-SCNN begins with a downsampling stage that reduces the input image to a fixed resolution of 360×640 pixels (width $\times$ height). To further reduce computational complexity, the number of convolutional channels is decreased. While the original Fast-SCNN employs 64 channels after downsampling, our modified version uses only 32 channels. This reduction effectively lowers the processing load, with a marginal trade-off in segmentation precision—an acceptable compromise for static and structured industrial environments.

2) *Global Feature Extraction*: Following the same principle, the global feature extraction stage also reduces the number of channels. In the original architecture, this stage uses 128 channels, whereas in the Reduced Fast-SCNN it is limited to 64 channels. Additionally, the pyramid pooling module is replaced with a lightweight average pooling layer. Compared to pyramid pooling, average pooling simplifies the computation and improves inference speed, making it more suitable for CPU-only real-time operation.

3) *Upsampling, Feature Fusion, and Classification*: The upsampling, feature fusion, and classification stages retain the core structure of the original Fast-SCNN but operate with fewer convolutional channels. The final output of the Reduced Fast-SCNN is a binary segmentation mask representing two classes: lane and non-lane regions. This segmentation output is later utilized to compute a confidence level that influences the vehicle's navigation decisions, particularly in adjusting steering and velocity control.

C. Integration of Both Systems

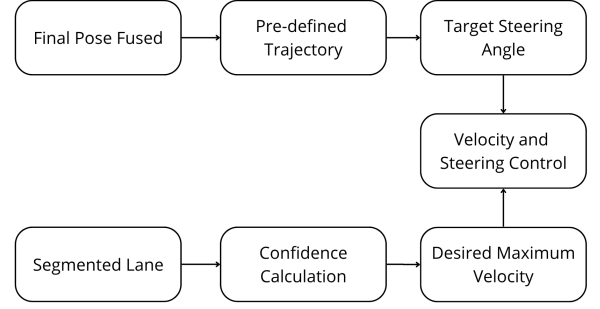


Fig. 3. Integration of the Extended RTAB-Map and Reduced Fast-SCNN for the navigation system.

Figure 3 presents the integrated architecture combining the Extended RTAB-Map for global localization and the Reduced Fast-SCNN for local perception. The navigation controller calculates the heading direction to the next waypoint as:

$$\theta_{\text{direction}} = \tan^{-1} \left(\frac{y_{\text{wp}} - y_{\text{position}}}{x_{\text{wp}} - x_{\text{position}}} \right) - \theta_{\text{orientation}} \quad (2)$$

where x_{wp} and y_{wp} are the coordinates of the next waypoint from the predefined trajectory, x_{position} and y_{position} represent the current vehicle position, and $\theta_{\text{orientation}}$ denotes the current vehicle heading. The resulting $\theta_{\text{direction}}$ is the desired heading angle toward the waypoint.

Using the Bicycle Model [11], the target steering angle is calculated as:

$$\delta_{\text{target}} = \tan^{-1} \left(\frac{2L \sin(\theta_{\text{direction}})}{D_{\text{lookahead}}} \right) \quad (3)$$

where L is the wheelbase (distance between front and rear axles), and $D_{\text{lookahead}}$ is the lookahead distance. The variable δ_{target} represents the desired steering angle for the vehicle.

Next, the system computes a confidence level from the segmented lane detection image. This confidence represents the proportion of detected lane area relative to a predefined region of interest (ROI) and is calculated as:

$$\text{normal} = \frac{\text{area of segmented lane detection}}{\text{area of ROI}} \quad (4)$$

where the numerator denotes the pixel area of the detected lane, and the denominator corresponds to the ROI area. The resulting normal value lies between 0 and 1.

The confidence value is then used to determine the target speed according to:

$$v_{\text{target_max}} = v_{\text{max}} \cdot \text{normal}^2 \quad (5)$$

where v_{max} is the maximum allowed velocity of the vehicle, and $v_{\text{target_max}}$ is the confidence-adjusted maximum speed. This ensures the vehicle moves more cautiously when lane confidence is low.

To ensure smooth transitions and suppress sudden speed fluctuations caused by segmentation noise or image flickering, a first-order low-pass filter is applied:

$$v_{\text{target}} = \alpha \cdot v_{\text{target_max}} + (1 - \alpha) \cdot v_{\text{target}} \quad (6)$$

where α is the smoothing coefficient (set to 0.055), $v_{\text{target_max}}$ is the instantaneous target speed, and v_{target} is the previously filtered speed. The final output, v_{target} , is used to control the vehicle's longitudinal motion, while δ_{target} regulates steering. Both values are transmitted to the low-level control layer to execute the motion commands in real time.

By combining these two subsystems, the navigation system achieves real-time perception and high-frequency localization, enabling stable and accurate motion control even on limited hardware resources.

III. RESULTS AND DISCUSSION

This section presents the experimental results obtained from the proposed navigation system. The evaluation focuses on three key aspects: (1) the accuracy of pose estimation with and without the Extended RTAB-Map, (2) the comparison of update rates between the default RTAB-Map and the Extended RTAB-Map, and (3) the inference performance comparison between the original Fast-SCNN and the proposed Reduced Fast-SCNN. All experiments were conducted on an Intel i5 12th Gen processor with 16 GB of RAM and no dedicated GPU.

A. Hardware Odometry Only vs. Extended RTAB-Map

The first experiment compared the pose estimation accuracy between the hardware odometry-only approach and the Extended RTAB-Map. The quantitative results are shown in Table I.

TABLE I
POSE ESTIMATION RESULT USING HARDWARE ODOMETRY ONLY

Lap-	X (m)	Y (m)	Error (m)
1	0.28	0.39	0.48
2	0.30	0.43	0.52
3	0.24	0.49	0.55
4	0.18	0.32	0.37
5	0.37	0.41	0.55
Average Error			0.494

In Table I, **Lap-** represents the lap number, while **X (m)** and **Y (m)** denote the final position of the vehicle (in meters) after traveling 500 meters. The **Error (m)** column shows the deviation of the estimated pose from the ground truth, which was defined as $(x = 0, y = 0)$. The average error of 0.494 m (49.4 cm) indicates that hardware odometry alone produces significant drift over time, rendering it unsuitable for precise autonomous navigation. Therefore, pose estimation improvement using the Extended RTAB-Map is essential.

The results obtained using the Extended RTAB-Map are presented in Table II.

TABLE II
POSE ESTIMATION RESULT USING EXTENDED RTAB-MAP

Lap-	X (m)	Y (m)	Error (m)
1	0.01	0.02	0.02
2	0.00	0.03	0.03
3	0.01	0.02	0.02
4	0.02	0.01	0.02
5	0.04	0.01	0.04
Average Error			0.026

The same parameters are used for Table II. After traveling 500 meters, the average positional error is 0.026 m (2.6 cm), demonstrating a significant improvement over the hardware odometry-only approach. This level of precision is sufficient for autonomous navigation in industrial environments, where centimeter-level accuracy is typically required.

Another way to visualize the accuracy improvement is through occupancy grid maps. Figures 4 and 5 compare the results between hardware odometry and the Extended RTAB-Map. In these maps, the blue line indicates the ground-truth trajectory.

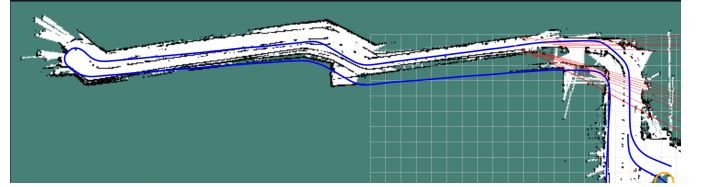


Fig. 4. Occupancy grid map generated using hardware odometry only.

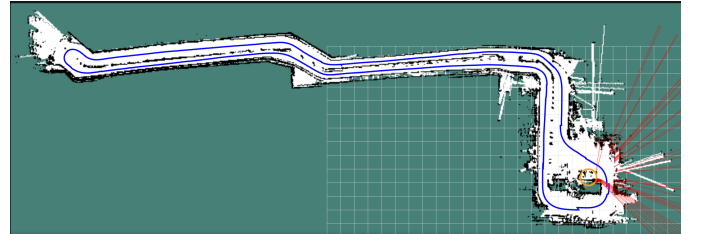


Fig. 5. Occupancy grid map generated using the Extended RTAB-Map.

Figures 4 and 5 clearly illustrate the difference between the two methods. The map produced by hardware odometry deviates substantially from the ground truth, exhibiting distorted features due to drift accumulation. In contrast, the map generated by the Extended RTAB-Map aligns closely with the ground truth trajectory, confirming the effectiveness of the proposed approach in enhancing pose estimation accuracy.

B. Update Rate: Default RTAB-Map vs. Extended RTAB-Map

The second experiment analyzed the update rate of the pose estimation between the default RTAB-Map and the Extended RTAB-Map. The measurement was conducted using the ROS topic statistics API, which computes the average publishing frequency of the pose topic. The results are shown in Table III.

TABLE III
UPDATE RATE COMPARISON BETWEEN DEFAULT AND EXTENDED
RTAB-MAP

Lap-	RTAB-Map (Hz)	Extended RTAB-Map (Hz)
1	1.003	50.029
2	1.002	50.006
3	1.223	49.999
4	1.056	49.999
5	1.332	49.998
Average	1.123	50.006

In Table III, **RTAB-Map (Hz)** represents the average frequency of the default RTAB-Map, and **Extended RTAB-Map (Hz)** represents the frequency achieved by the proposed system. The average update rate increased from 1.123 Hz to 50.006 Hz, a nearly fiftyfold improvement. This demonstrates that integrating hardware odometry through the proposed skip connection and Kalman fusion significantly enhances temporal resolution, making real-time pose updates possible without requiring GPU acceleration.

C. Reduced Fast-SCNN Result

The third experiment evaluated the performance of the proposed Reduced Fast-SCNN model. A custom dataset of 797 labeled images with a resolution of $640 \times 360 \times 3$ was developed specifically for this task. The dataset was not divided into training and validation subsets because it represents a complete and absolute sample of the target environment. The model was trained using the Adam optimizer with a learning rate of 0.0001, a batch size of 4, and 100 epochs. A sample of the dataset is shown in Fig. 6.

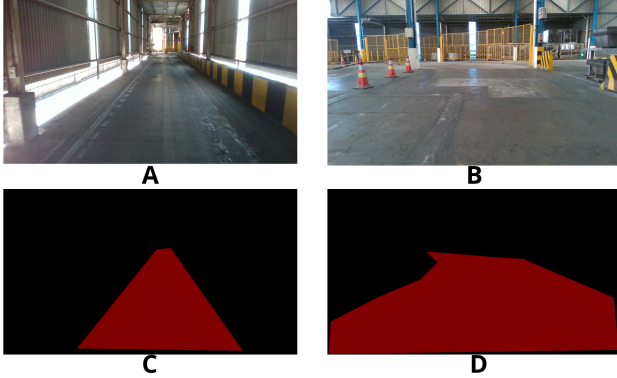


Fig. 6. Samples from the custom dataset used for training the Reduced Fast-SCNN.

In Fig. 6, images A and B represent RGB inputs from the onboard camera, while images C and D show the corresponding segmentation masks. The segmentation results contain two classes: lane and non-lane regions. The lane pixels are marked in red, while the background is represented in black.

After training, the model was converted into the ONNX format for deployment on CPU hardware. Inference was performed using the ONNX Runtime, which provided approximately a twofold increase in processing speed compared to the native PyTorch implementation [12]. Figure 7 shows

preliminary results of lane detection using the Reduced Fast-SCNN.

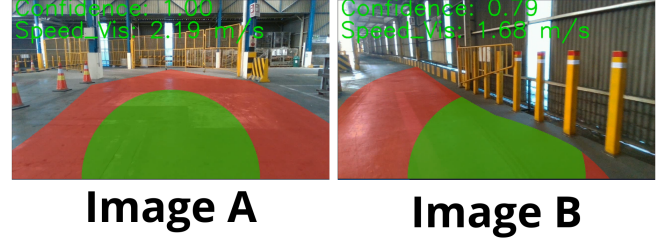


Fig. 7. Preliminary lane detection results using the Reduced Fast-SCNN.

In Fig. 7, the red mask indicates the detected lane area, while the green region represents the region of interest (ROI) used to calculate the confidence level. Image A shows a full straight lane where the ROI is completely filled, while Image B shows a turning condition with a truncated ROI, resulting in lower confidence.

The final inference results are displayed in Fig. 8.

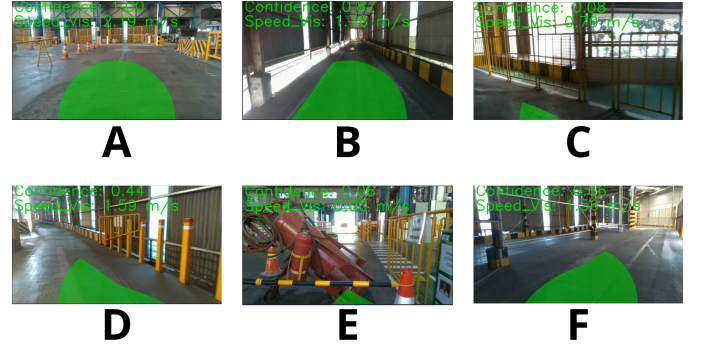


Fig. 8. Lane detection and velocity control results using the Reduced Fast-SCNN.

Figure 8 illustrates various driving scenarios. In Image A, the vehicle is in a full straight lane with an output velocity of 2.2 m/s. Image B depicts a narrow straight lane with a slightly reduced speed of 1.76 m/s. In Image C, the vehicle navigates a sharp turn while approaching a fence, resulting in a velocity of 0.70 m/s. Image D shows a left turn in a narrow area, where the velocity is 1.59 m/s. In Image E, the vehicle encounters an obstacle directly in front, stopping completely at 0.00 m/s. Finally, Image F depicts a right turn in a narrow area with a speed of 1.56 m/s. These results demonstrate the effectiveness of the Reduced Fast-SCNN in adjusting the vehicle's speed based on lane confidence and environmental conditions.

D. Original Fast-SCNN vs. Reduced Fast-SCNN Inference Time

The final experiment compared the inference time between the original Fast-SCNN and the proposed Reduced Fast-SCNN models. Both were tested on the same hardware platform (Intel i5 12th Gen CPU, 16 GB RAM, Intel Iris Xe Graphics). The same test images with a resolution of $640 \times 360 \times 3$ were used for consistency. The results are summarized in Table IV.

TABLE IV
COMPARISON OF INFERENCE TIME BETWEEN ORIGINAL AND REDUCED FAST-SCNN

Frame Number	Fast-SCNN (ms)	Reduced Fast-SCNN (ms)
1	22.4	7.2
2	29.3	7.7
3	34.2	7.8
4	39.2	7.7
5	23.8	7.4
6	28.1	8.0
Average	29.5	7.63

Table IV clearly indicates that the Reduced Fast-SCNN significantly outperforms the original version in terms of speed. The average inference time of the original Fast-SCNN is 29.5 ms per frame, while the Reduced Fast-SCNN achieves an average of 7.63 ms. This improvement corresponds to a nearly $3.9\times$ reduction in computation time. The results confirm that the proposed architecture is capable of real-time lane detection using CPU-only hardware, making it well suited for industrial autonomous vehicle applications where computational resources are limited.

IV. FUTURE DEVELOPMENT

There are two primary avenues to improve the local navigation system. The first is to increase robustness in lane detection. This can be achieved by incorporating temporal modeling, for example by using gated recurrent units (GRUs) [13] to exploit inter-frame dependencies and reduce flicker in the segmentation output. The second direction is to perform velocity and steering control using machine learning. In this approach, control commands can be learned by leveraging the outputs of the lane detection module together with the fused pose estimation, enabling adaptive policies that respond to confidence and environmental context.

V. CONCLUSION

By utilizing and extending RTAB-Map as the primary algorithm for vehicle pose estimation, we achieve centimeter-level accuracy (error < 5 cm). This is enabled by introducing a skip connection from hardware odometry to the pose fusion stage. During fusion, a Kalman Filter combines the corrected pose from RTAB-Map with the odometry-based state, yielding smooth and accurate estimates. The final fused pose attains a 50 Hz update rate on an Intel i5 12th Gen system with 16 GB RAM and no dedicated GPU.

On the perception side, reducing convolutional channels and replacing pyramid pooling with average pooling yields

a lightweight, fast segmentation network that runs on CPU alone with sub-10 ms inference. The proposed Reduced Fast-SCNN attains an average inference time of 7.63 ms on the same hardware platform.

By combining the Extended RTAB-Map and the Reduced Fast-SCNN, we realize a CPU-only navigation system that delivers high-accuracy, high-frequency pose estimation together with real-time lane detection—well suited for industrial autonomous vehicles operating under compute and latency constraints.

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